

PlanTrace-VL: auditable vision-language grounding for document-centered planning review

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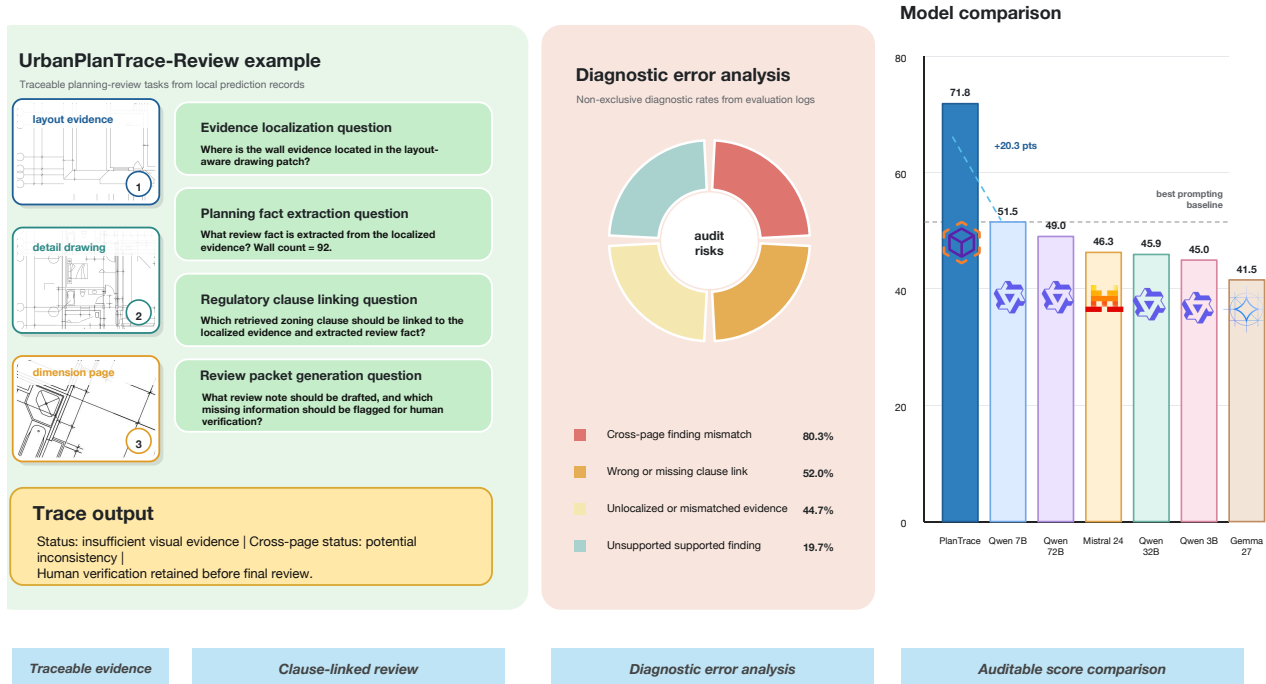


Figure 1: **The UrbanPlanTrace-Review benchmark.** **Left:** Traceable tasks connect document evidence to planning facts and candidate regulatory clauses. **Middle:** Diagnostic errors expose evidence-linking and cross-page risks. **Right:** Auditable review scores compare PlanTrace-VL with prompting baselines. Model-size labels use B for billions of parameters, and pts denotes percentage points.

Abstract

Professional planning review must connect findings to evidence across multi-page maps, drawings, annotations, and regulatory text, yet answer-oriented vision-language approaches leave this evidence-to-rule chain difficult to inspect. We present PlanTrace-VL, a document-centered, human-in-the-loop framework that decomposes review into evidence grounding, planning-fact extraction, regulatory-clause retrieval and linking, structured review-packet generation, and verification status. UrbanPlanTrace-Review provides a curated open-source proxy benchmark for these traceable intermediate outputs, and two-stage low-rank adaptation fine-tuning adapts the model to technical drawings and regulatory text. Across four 300-sample diagnostic tasks, PlanTrace-VL achieves 95.67% Evidence Localization and a 92.89% Fact F1 score, where F1 is the harmonic mean of precision and recall. However, the Consistency F1 score remains 19.66% and the unsupported-link rate is 19.67%, exposing unresolved cross-page and regulatory-grounding risks. PlanTrace-VL therefore supports auditable review assistance, not autonomous approval: final interpretation, compliance judgment, and approval remain with human planners.

Keywords

Vision-language models, Urban planning document review, Human-artificial intelligence collaboration, Urban planning, Multimodal evidence grounding

Introduction

Computer-based planning tools have long been framed as systems that support, rather than replace, professional reasoning (Klosterman, 1997; Geertman and Stillwell, 2004). Recent language-model research extends this tradition to plan reading, policy analysis, public participation, and planning decision support (Peng et al., 2024; Fu, 2024; Fu et al., 2024). Professional planning review, however, poses a particularly demanding support problem. The unit of analysis is often a multimodal case packet containing zoning maps, site plans, permit drawings, legends, annotations, dimensions, frontage and parcel information, and regulatory excerpts. A defensible finding must therefore remain connected to the document evidence and regulatory provision on which it rests.

Visually rich document understanding provides useful foundations for this setting. The Document Visual Question Answering (DocVQA) benchmark established visual question answering (VQA) over document images, while the Document Understanding Dataset and Evaluation (DUDE) and Global Reasoning for Multi-Page VQA (GRAM) ex-

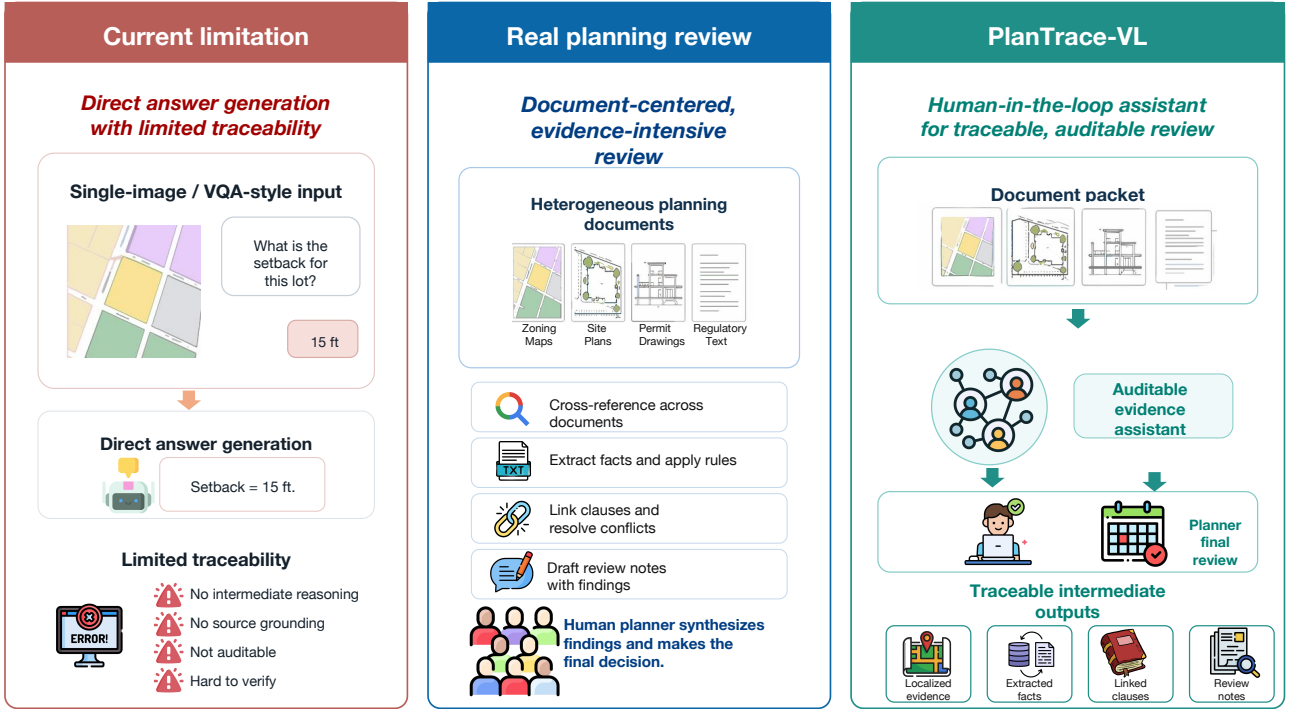


Figure 2: Conceptual motivation of PlanTrace-VL. VQA denotes visual question answering, and ft denotes feet.

tended evaluation and modeling toward diverse, multi-page documents (Mathew et al., 2021; Van Landeghem et al., 2023; Blau et al., 2024). Domain-specific vision-language models have also advanced planning-map interpretation (Zhu et al., 2025). Yet answer-oriented document and map benchmarks do not by themselves represent the evidentiary workflow of planning review: locating the relevant region, extracting a planning fact, linking that fact to a candidate clause, recording whether the link is supported, and flagging the result for human verification. Automated compliance-checking research similarly shows that regulatory review requires explicit representations of both design information and rules (Solihin and Eastman, 2015; Zhang and El-Gohary, 2016, 2017). The open problem is thus not simply multi-page question answering, but auditable grounding across heterogeneous planning evidence and regulatory text.

Document-scale assistance also requires evidence selection before generation. Textual retrieval-augmented generation (RAG) and dense passage retrieval separate candidate-source retrieval from response generation, partly addressing provenance and updateability (Lewis et al., 2020; Karpukhin et al., 2020). Visual retrieval systems such as ColPali and VisRAG instead retrieve page images directly to preserve layout, figures, and tables (Faysse et al., 2025; Yu et al., 2025). Long-document benchmarks further show that evidence may be distributed across pages or documents and that some questions are deliberately unanswerable (Ma et al., 2024; Cho et al., 2025). In planning review, however, selecting a relevant page is only the start: the system must preserve how a region, extracted fact, and candidate clause support each note.

Reliability makes this distinction consequential. Large

vision-language models (VLMs) can generate image-inconsistent objects and conclusions, and dedicated benchmarks continue to expose visual hallucination and image-context reasoning failures (Li et al., 2023; Guan et al., 2024). A fluent review note can therefore appear resolved even when its evidence or regulatory link is weak. Research on interaction between humans and artificial intelligence (AI) systems recommends interfaces that preserve user control, communicate uncertainty, and discourage uncritical reliance (Amershi et al., 2019; Bućinca et al., 2021; Mosqueira-Rey et al., 2023). For planning review, these principles imply a concrete technical requirement: the model should expose inspectable intermediate artifacts instead of hiding them behind a final answer. Figures 1 and 2 summarize this shift from answer generation to document-centered, auditable review.

To meet this requirement, we introduce *PlanTrace-VL*, a human-in-the-loop framework for auditable vision-language grounding in planning document review. PlanTrace-VL treats the model as a review-support system that exposes evidence rather than makes final decisions. It localizes visual evidence in maps and drawings, extracts planning facts such as land-use labels and dimensional notes, links those facts to candidate regulatory clauses, and drafts review notes that separate supported findings from cases requiring human verification. Its scope is deliberately bounded to review support; automated approval, complete legal compliance checking, and professional judgment remain outside the system’s claims.

We also introduce UrbanPlanTrace-Review, a curated open-source proxy benchmark and task suite for training and evaluating this workflow. The benchmark covers visual grounding, regulatory clause linking, review-packet genera-

tion, and cross-page consistency. PlanTrace-VL uses a two-stage supervised fine-tuning (SFT) strategy based on low-rank adaptation (LoRA): technical-drawing adaptation first, followed by regulatory-clause-grounded instruction tuning. At inference time, layout-aware patch extraction, planning fact extraction, hierarchical clause retrieval, clause-link consistency checks, and structured review-note generation produce a trace that a planner can inspect.

The evaluation measures each layer of this trace rather than only final-answer accuracy. On four 300-sample diagnostic tasks, PlanTrace-VL reaches 95.67% Evidence Localization and a 92.89% Fact F1 score, where F1 is the harmonic mean of precision and recall; these are the strongest results in the reported comparison. The remaining results also delimit the contribution: the Consistency F1 score is only 19.66%, the unsupported-link rate is 19.67%, and the model does not reliably distinguish insufficient visual evidence from other verification cases. These outcomes support a focused claim—domain adaptation improves several traceable intermediate outputs—without establishing municipal deployment readiness or autonomous review capability.

This paper makes three contributions. First, it formulates planning document review as an auditable evidence-grounding workflow rather than a single-answer prediction task. Second, it introduces UrbanPlanTrace-Review and a two-stage training pipeline that connects technical-drawing understanding with regulatory-clause-grounded review. Third, it provides a layer-wise diagnostic protocol that identifies both improvements in traceable intermediate outputs and failure modes that still require planner oversight.

Related Work

AI-Assisted Planning and Urban Intelligence

Planning support systems traditionally combine data, models, and visualization to assist professional planning activities (Klosterman, 1997; Geertman and Stillwell, 2004). Contemporary planning support science further emphasizes demand-driven tools, governance context, and transparent integration into planning processes rather than technology-led deployment (Geertman and Stillwell, 2020). Language models extend this agenda to plan reading, policy synthesis, public participation, and decision assistance, while retaining concerns about professional judgment, public accountability, and institutional context (Peng et al., 2024). Natural language processing (NLP) studies analyze resilience plans, general plans, and policy documents at scale to reveal themes, omissions, and jurisdictional variation (Fu et al., 2023; Brinkley and Stahmer, 2024; Fu, 2024). More recent work tests whether general-purpose large language models (LLMs) can evaluate plans or support participatory workflows (Fu et al., 2024; Zhou et al., 2024b). These studies establish the value of language-based planning assistance, but they primarily analyze text or planning knowledge. PlanTrace-VL instead targets review packets in which visual evidence and regulatory text must be connected through an inspectable trace.

Urban intelligence benchmarks. Urban planning and urban

computing benchmarks test whether foundation models can reason about cities as social, spatial, and temporal systems. UrbanPlanBench evaluates planning principles, professional knowledge, and regulations (Zheng et al., 2025); CityBench and USTBench address broader urban reasoning, world modeling, and spatio-temporal prediction (Feng et al., 2024; Lai et al., 2025). Agent-oriented platforms additionally study how language-model agents act in or reason about simulated city environments (Xu et al., 2023). These benchmarks move beyond generic NLP, but their principal targets remain planning knowledge, urban reasoning, prediction, or agent behavior. UrbanPlanTrace-Review evaluates a different workflow: finding document evidence, extracting review-relevant facts, linking them to candidate clauses, and preserving a trace for planner verification.

Multi-Page Documents and Technical Drawings

Document AI provides the closest general formulation of the paper’s multimodal input problem. DocVQA established visual question answering over document images and highlighted the importance of document structure (Mathew et al., 2021). DUDE broadened this setting to diverse, multi-domain, and multi-page documents, while GRAM introduced document-level reasoning across pages (Van Landeghem et al., 2023; Blau et al., 2024). This literature demonstrates that text, layout, and visual content must be modeled jointly and that page-level processing is insufficient for many real documents. Its typical output, however, remains an answer to a question. PlanTrace-VL adopts multi-page document reasoning as a foundation but evaluates a planning-specific chain of localized evidence, extracted facts, candidate regulatory clauses, support status, and review notes.

Multi-page evidence selection. Multi-page DocVQA (MP-DocVQA) uses hierarchical multimodal encoding to identify relevant pages and generate answers (Tito et al., 2023). MMLongBench-Doc evaluates long portable document format files whose answers depend on text, images, charts, tables, and page locations, including cross-page and unanswerable cases (Ma et al., 2024); M3DocVQA, a benchmark for multimodal, multi-page, multi-document understanding, further distributes evidence across multiple documents (Cho et al., 2025). Retrieval-centered methods separate evidence selection from generation: dense retrieval and RAG surface candidate passages (Karpukhin et al., 2020; Lewis et al., 2020), whereas ColPali and VisRAG index page images to preserve visual structure (Faysse et al., 2025; Yu et al., 2025). These approaches make source selection more explicit, but they generally evaluate retrieval or answer quality rather than a planning trace linking visual evidence to regulatory interpretation and verification status.

Vision-Language Models for Urban Materials

Vision-language research provides multimodal foundations for interpreting maps, remote-sensing imagery, urban scenes, and planning materials. Remote Sensing Visual Question Answering (RSVQA), GeoChat, RemoteCLIP, EarthGPT, and GEOBench-VLM evaluate geospatial ques-

tion answering, image–text alignment, localization, and spatial interpretation (Lobry et al., 2020; Kuckreja et al., 2023; Liu et al., 2023; Zhang et al., 2024; Danish et al., 2025). UrBench extends multimodal evaluation to multi-view urban scenarios (Zhou et al., 2024a). Most directly, PlanGPT-VL adapts vision-language modeling to planning-map interpretation and uses structured verification to reduce hallucination (Zhu et al., 2025). PlanTrace-VL builds on this domain-specific direction but changes the unit of evaluation from a map answer to an evidence chain spanning documents, facts, candidate clauses, and review status.

Technical drawing resources. Planning drawings differ from natural images because symbols, linework, scales, labels, legends, and dimensions jointly encode design meaning. FloorPlanCAD and ArchCAD-400K provide large-scale resources for panoptic symbol spotting in computer-aided design (CAD) drawings, while CubiCasa5K supports recognition of rooms, walls, openings, and related floor-plan elements (Fan et al., 2021; Luo et al., 2025; Kalervo et al., 2019). Specialized symbol-spotting research likewise shows that architectural semantics require drawing-aware visual grounding (Rezvanifar et al., 2020). TechMB extends this concern to vision-language interpretation of technical drawings (Kunz et al., 2025). These resources motivate PlanTrace-VL’s first-stage visual adaptation. They do not address the subsequent planning-review problem of converting localized evidence into facts, regulatory links, and verification-aware notes.

Regulatory Grounding and Human-Centered AI

Planning document review is also a regulatory interpretation task. Land-use labels, frontage conditions, setbacks, height limits, lot coverage, overlays, and parcel boundaries matter because zoning and development codes translate planning goals into review criteria (Talen, 2012). Automated compliance checking represents rules and design information computationally through rule classification, natural-language information extraction, logic-based reasoning, building information modeling-based checking, and machine-readable workflows (Dimyadi and Amor, 2013; Eastman et al., 2009; Solihin and Eastman, 2015; Zhang and El-Gohary, 2016, 2017; Demir Altintas and Ilal, 2022). Adoption research nevertheless identifies technical, commercial, political, and governance barriers and retains a role for human oversight (Beach et al., 2020). CODE-ACCORD, the National Zoning and Land Use Database (NZLUD), and the National Zoning Atlas further illustrate the growing availability of structured or semi-structured regulatory data (Hettiarachchi et al., 2025; Mleczko and Desmond, 2023; National Zoning Atlas, 2026). PlanTrace-VL uses these resources to surface candidate provisions and expose the evidence behind a link. It does not claim autonomous planning approval or definitive legal compliance checking.

Human-centered auditability. Because planning review affects development rights and public accountability, AI assistance must be reviewable rather than merely fluent. Human–AI interaction research emphasizes user control, uncertainty communication, and appropriate reliance (Amershi et al., 2019; Mosqueira-Rey et al., 2023). Reviewable automated

decision-making similarly calls for records that can be inspected, contested, and governed (Cobbe et al., 2021), while evaluative AI emphasizes evidence for and against a decision rather than a one-shot recommendation (Miller, 2023; Le et al., 2024). This need is reinforced by documented hallucination and visual-reasoning failures in large vision-language models (Li et al., 2023; Guan et al., 2024). Visual contrastive decoding can reduce object hallucination, yet free-form hallucination benchmarks show that improvement on one error formulation need not transfer to another (Leng et al., 2024; Kaul et al., 2024). Evidence that uncritical reliance can persist in AI-assisted decisions adds a distinct human-factors risk (Bućinca et al., 2021). PlanTrace-VL operationalizes these principles through evidence localization, fact extraction, clause relevance, unsupported-link detection, review-note factuality, human-verification behavior, and cross-page consistency.

Together, prior work advances planning LLMs, urban benchmarks, multi-page document understanding, map-oriented VLMs, drawing interpretation, regulatory rule representation, and human-centered AI. PlanTrace-VL connects these strands around a narrower problem: producing an auditable planning-review trace from document evidence to planner-verifiable notes. The methodology below translates this synthesis into a proxy benchmark, training strategy, inference pipeline, and diagnostic evaluation.

Methodology

Review Formulation

PlanTrace-VL treats planning document review as a review-packet task rather than a single-image visual question-answering task. In practice, a planner may inspect zoning maps, site plans, permit drawings, legends, annotations, dimensional notes, frontage information, parcel boundaries, and regulatory excerpts as one evidentiary record. The methodological objective is to generate traceable intermediate artifacts that a planner can inspect, compare, and revise, not to produce an autonomous approval decision. This design principle determines the rest of the pipeline: the benchmark must contain review-relevant evidence, the tasks must expose intermediate outputs, and the inference schema must preserve uncertainty. PlanTrace-VL is therefore organized around auditable grounding, where each review note should be traceable to localized visual evidence, extracted planning facts, and candidate regulatory clauses.

Benchmark and Task Design

Real municipal permit-review packets are often sensitive, inconsistent, and not publicly redistributable. We therefore construct UrbanPlanTrace-Review as a curated open-source proxy benchmark and task suite for auditable planning document review. The benchmark combines public architectural drawings, technical diagrams, floor-plan resources, and structured regulatory text to approximate review-relevant visual and textual signals. This construction supports reproducibility and allows the data and task design to be inspected. It also defines a central boundary of the study: UrbanPlanTrace-Review is a proxy benchmark for studying auditable planning review, not a complete representation

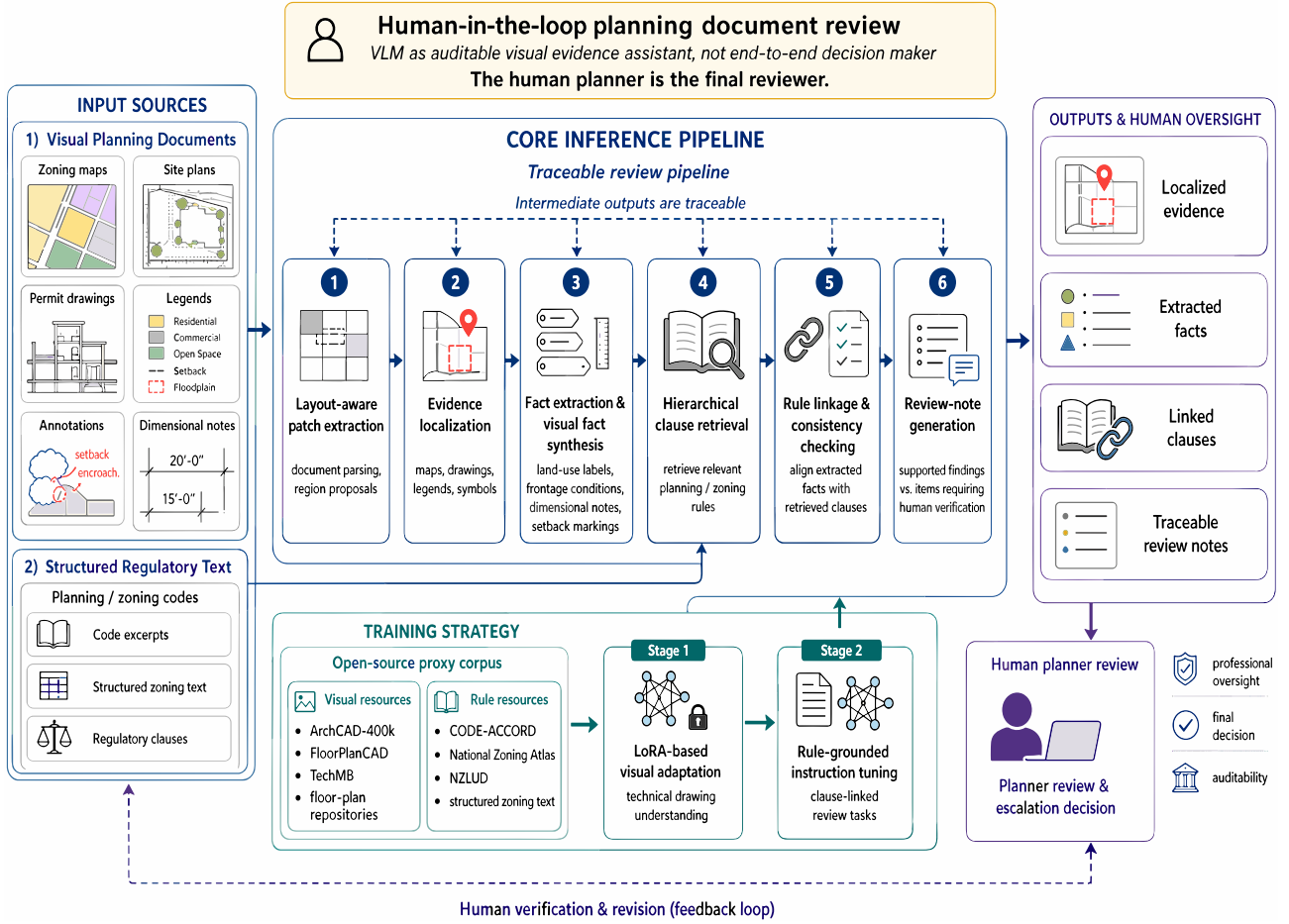


Figure 3: Overview of the PlanTrace-VL framework.

of municipal permit-review practice, local administrative discretion, or official legal review records. Because its visual resources include substantial technical drawing and CAD-style material, the benchmark likely overrepresents drawing-element localization relative to full urban planning review packets. It should therefore be interpreted as a controlled diagnostic setting for traceable review behavior, not as evidence of municipal deployment readiness.

Figure 4 summarizes UrbanPlanTrace-Review at the corpus level. The distribution separates visual adaptation, regulatory-clause-grounded review tuning, and diagnostic review tasks. The accompanying statistics report global corpus properties that complement, rather than duplicate, the task-split table.

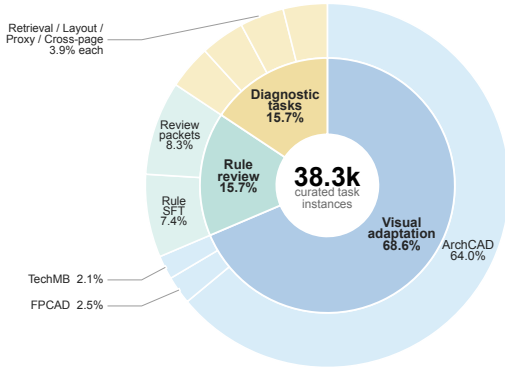
The corpus-level summary is complemented by sample-level examples in Figure 5. These examples clarify how UrbanPlanTrace-Review differs from a conventional visual question-answering benchmark. Each record is organized around an auditable review step: a document input, a review-oriented question, a gold structured output, and a trace note that preserves the need for planner verification.

Visual sources. UrbanPlanTrace-Review draws its visual-grounding data from ArchCAD-400k, FloorPlanCAD, TechMB, and related public floor-plan repositories. These resources expose the model to dense technical materials with symbols, legends, annotations, structural elements,

room or object labels, dimensional markings, and layout patches. Such elements provide candidate evidence for planning facts, including labeled uses, frontage conditions, setback-related markings, dimension notes, and regions requiring human checking.

Rule-grounding resources. The benchmark incorporates CODE-ACCORD, the National Zoning Atlas, NZLUD, and structured zoning text. These sources provide regulatory clauses and structured land-use information for retrieval and clause-linking tasks. Retrieved provisions remain candidates for review rather than final legal determinations, and the benchmark does not capture all local exceptions, variances, administrative interpretations, or discretionary procedures.

Task design. UrbanPlanTrace-Review converts planning review into three linked groups of inspectable tasks. First, evidence localization identifies where a planner should look in a map, drawing, diagram, or layout patch, and planning fact extraction converts that localized evidence into structured facts such as land-use labels, frontage conditions, dimensional notes, setback-related markings, or annotations. Second, regulatory clause linking ranks candidate provisions for an extracted fact, while review-packet generation organizes evidence, facts, and clauses into a structured note that separates supported findings from uncertain or unsupported statements. Third, cross-page consistency checking



Dataset statistics

Statistic	Value
Curated task instances	38,291
Visual source families	3
Diagnostic task families	4
Evaluation samples per diagnostic task	300
Indexed zoning/code records	9,677
Retrieval candidates per sample	8
Layout evidence-label classes	26
Proxy visual-topic classes	3
Cross-page status classes	3

Data sources: ArchCAD, FloorPlanCAD, TechMB; CODE-ACCORD, National Zoning Atlas, National Zoning and Land Use Database (NZLUD), and structured zoning text.

Figure 4: UrbanPlanTrace-Review dataset overview. The ring chart shows the distribution of curated task instances used to construct the benchmark, and the table reports global corpus statistics that complement the detailed train/evaluation split in Table 1. Percentages in the ring chart are computed over task instances; zoning/code records are reported separately because they form the retrieval index rather than supervised task examples. SFT denotes supervised fine-tuning, and FPCAD abbreviates FloorPlanCAD.

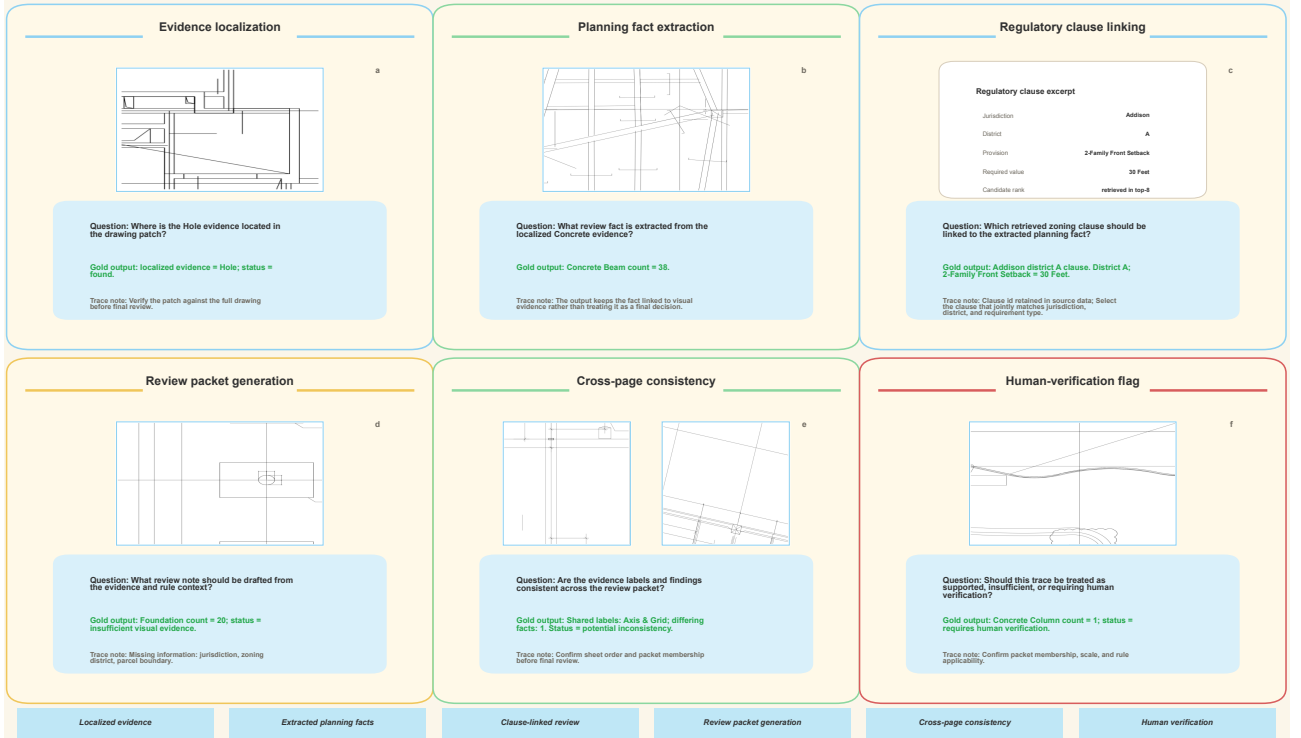


Figure 5: Representative UrbanPlanTrace-Review samples across auditable planning-review tasks. The examples illustrate evidence localization, planning fact extraction, regulatory clause linking, review-packet generation, cross-page consistency checking, and human-verification status prediction. Each panel is drawn from logged task records. Green text marks gold structured outputs, not final planning approvals.

tests whether facts, labels, notes, and clause links agree across documents, and human-verification status prediction marks outputs as supported, uncertain, or requiring planner review. Table 1 reports the corresponding train, validation, and evaluation counts.

Two-Stage Model Adaptation

Training follows a two-stage LoRA-based supervised fine-tuning strategy. Stage 1 performs visual adaptation for

technical drawing understanding. In this stage, the model is tuned to parse floor plans, drawing symbols, legends, annotations, structural elements, layout patches, and dimensional markings. This stage addresses the visual conventions of planning-related drawings, which differ from natural images and generic VQA benchmarks.

Stage 2 performs regulatory-clause-grounded instruction tuning for regulatory-clause-linked review tasks. In this

stage, the model is trained on examples that combine localized evidence, extracted planning facts, candidate clauses, review-note structures, and human-verification labels. The stage order is deliberate: the model first learns to identify technical visual evidence, then learns to connect that evidence to regulatory clauses and review notes. This sequence is intended to reduce the risk that outputs involving regulatory clauses are generated without stable visual support.

Inference, Auditability, and Scope

Figure 3 summarizes the inference workflow. First, layout-aware patch extraction identifies document regions likely to contain review-relevant evidence. Second, evidence localization assigns candidate evidence descriptions or region labels to those patches. Third, planning fact extraction converts localized evidence into structured planning facts. Fourth, hierarchical clause retrieval searches the regulatory text index to surface candidate provisions at the corpus, document, and clause level. Fifth, consistency checking for regulatory clause linking compares extracted facts with retrieved clauses and flags weak or mismatched links. Sixth, review-note generation drafts a structured note that records localized evidence, extracted facts, linked regulatory clauses, and verification status. Finally, a human planner reviews the trace, accepts or revises supported findings, and escalates uncertain or unsupported items. The model generates candidate evidence chains and notes; interpretation, escalation, and professional judgment remain under planner oversight.

Formal review trace. We write the construction of one review trace as

$$\begin{aligned} E &= g_\theta(D), & F &= h_\theta(E), \\ C_K &= \mathcal{R}_C(F; K), \\ (Y, S) &= q_\theta(D, E, F, C_K), \\ \tau &= \langle D, E, F, C_K, Y, S \rangle. \end{aligned} \quad (1)$$

In Eq. (1), D is the input packet; E is the localized-evidence set; F is the extracted-fact set; C_K is the set of the top- K retrieved clauses; Y is the generated-finding set; and S is the verification status. The functions g_θ , h_θ , and q_θ denote evidence localization, fact extraction, and review generation under model parameters θ . The operator $\mathcal{R}_C(\cdot; K)$ retrieves K clauses from the indexed corpus C , and τ denotes the resulting ordered review record. For any supported finding $y \in Y_{\text{sup}}$, structural traceability requires an exposed path $e \rightarrow f \rightarrow c \rightarrow y$ with $e \in E$, $f \in F$, and $c \in C_K$. The symbol $\in$ denotes set membership, while $\rightarrow$ denotes a claimed support link rather than guaranteed correctness; link correctness is evaluated empirically.

Output schema. PlanTrace-VL defines auditability through the structure of its outputs. Each generated review trace is expected to contain localized evidence, extracted facts, linked regulatory clauses, supported findings, unsupported or uncertain findings, human-verification flags, and review notes. These fields correspond to review actions in professional planning practice: locating evidence, checking factual claims, comparing a claim with applicable stan-

dards, identifying unsupported links, and deciding whether a case requires further human review. This schema is more inspectable than a single final answer because it allows planners to follow the reasoning path, identify where an error may have entered, and preserve an accountable review record. It also determines how the experiments are evaluated, since each metric targets one layer of the trace rather than a single end label.

Scope. This study uses UrbanPlanTrace-Review to examine auditable evidence grounding in planning document review. UrbanPlanTrace-Review is a curated open-source proxy benchmark and task suite rather than a municipal permit-review corpus. PlanTrace-VL does not claim to automate official planning approval, replace professional judgment, or fully replicate municipal review practice. It should be understood as a research framework that generates traceable intermediate outputs for planner verification.

Experiments

Evaluation Setup

The experiments evaluate PlanTrace-VL as a diagnostic system for auditable planning document review. The evaluation target is review support, not autonomous approval or rejection of planning documents. It follows the review sequence that a planner would need to inspect: localizing visual evidence, extracting planning facts, linking those facts to regulatory clauses, generating structured review notes, preserving human-verification flags, and checking cross-page consistency. Each experiment therefore corresponds to a review-support step. Visual grounding tests whether evidence can be traced to a document region. Planning fact extraction tests whether drawings and annotations can be converted into reviewable statements. Regulatory clause linking tests whether those statements can be connected to candidate regulatory provisions. Review-packet generation tests whether the output remains parseable, factual, and appropriately cautious. The results are interpreted as evidence about auditable review support, not as a generic model leaderboard.

Tasks and data splits. Table 1 reports the UrbanPlanTrace-Review task construction and data splits used in the reported PlanTrace-VL experiments. Each split corresponds to a review action in document-centered planning practice. Stage 1 visual SFT supports technical drawing adaptation, so the model can interpret dense visual signals such as symbols, layout patches, legends, annotations, and dimensional markings. Stage 2 rule/review SFT supports regulatory-clause-grounded review tuning, where extracted planning facts must be connected to regulatory text and expressed as inspectable review notes. The rule corpus index provides candidate zoning and code provisions for retrieval. The retrieval rule split tests whether the relevant provision can be surfaced before generation. The layout patch grounding split evaluates whether visual evidence can be localized and converted into planning facts. The review packet proxy split evaluates structured review-note generation. The cross-page consistency split tests whether distributed evidence can be reconciled across a proxy packet.

Table 1: UrbanPlanTrace-Review task construction used in the reported PlanTrace-VL experiments. Counts are taken from the local UrbanPlanTrace-Review manifests and training configurations.

Task/data split	Main source signal	Training / pool	Evaluation / validation	Split source	Supported evaluation
Stage 1 visual SFT	ArchCAD / FloorPlanCAD / TechMB	26,270 pool	789 validation	runtime split	Technical drawing adaptation
Stage 2 rule/review SFT	Structured zoning text + review packets	6,021 pool	302 validation	runtime split	Regulatory-clause-grounded review tuning
Rule corpus index	Zoning and code text	9,677 documents	Not applicable	retrieval index	Candidate rule retrieval
Retrieval rule	Rule candidates + gold clause	1,200 training	300 evaluation	manifest split	Clause retrieval
Layout patch grounding	Drawing/map patch + localized labels	1,200 training	300 evaluation	manifest split	Evidence localization, planning fact extraction
Review packet proxy	Image/facts + candidate clauses	1,200 training	300 evaluation	manifest split	Review packet quality
Cross-page consistency	Multi-page proxy packet	1,200 training	300 evaluation	manifest split	Cross-page consistency

Baselines and fairness. Table 2 compares PlanTrace-VL with prompting-based baselines. The evaluated suite covers Mistral-Small-3.2-24B-Instruct-2506 (Mistral AI, 2025; Mistral AI Team, 2025), MiniCPM-V-2.6-8B (Yao et al., 2024), Gemma-3-27B/12B/4B-IT (Gemma Team et al., 2025), Qwen3-VL-32B/30B-A3B/8B/4B-Instruct (Bai et al., 2025b), and Qwen2.5-VL-72B/32B/7B/3B-Instruct (Bai et al., 2025a). In these model names, B denotes billions of parameters, IT denotes an instruction-tuned variant, and A3B denotes approximately 3 billion activated parameters. The comparison is diagnostic rather than leaderboard-oriented. It tests whether domain adaptation and structured outputs improve auditability for planning review, not whether the largest model has the strongest generic visual question-answering ability. This distinction matters because a model may answer an isolated clause-selection question correctly while failing to expose localized evidence, trace fields, uncertainty status, or review notes that a planner can verify. For transparency, the baseline run produced all 52 expected prediction files with 300 rows per file and no malformed records in JSON Lines, a newline-delimited JavaScript Object Notation (JSON) format; the aggregate is still marked incomplete because it contains 15 sample-level application programming interface errors. The PlanTrace-VL evaluation is complete with 0 errors across the four scored diagnostic tasks.

Auditability metrics. We group the metrics into auditability layers rather than treating them as a single accuracy score. Some metrics are reported as headline component scores, while others appear in diagnostic tables or as source metrics for error analysis. Their per-sample and aggregate definitions are

$$\begin{aligned}
 P_i &= \frac{TP_i}{TP_i + FP_i}, & R_i &= \frac{TP_i}{TP_i + FN_i}, \\
 F_{1,i} &= \frac{2P_iR_i}{P_i + R_i}, & \bar{F}_1 &= \frac{1}{N} \sum_{i=1}^N F_{1,i}, \\
 \text{Acc} &= \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_i = y_i], & \text{FSR} = \text{ULR} &= \frac{1}{N} \sum_{i=1}^N u_i, \\
 \text{R@}k &= \frac{1}{N} \sum_{i=1}^N \mathbf{1}[r_i \leq k], & \text{MRR} &= \frac{1}{N} \sum_{i=1}^N \frac{1}{r_i}.
 \end{aligned} \tag{2}$$

In Eq. (2), i indexes one of N evaluation samples. The quantities TP_i , FP_i , and FN_i are the true-positive, false-positive, and false-negative counts for sample i ; P_i , R_i , and $F_{1,i}$ are its precision, recall, and F1 score; and $\bar{F}_1$ is macro-F1. Evidence-label F1 uses set overlap, whereas

token F1 uses lowercased, whitespace-normalized token multisets so repeated tokens retain their counts. We set $F_{1,i} = 1$ when both sides are empty and $F_{1,i} = 0$ when exactly one side is empty.

The labels y_i and $\hat{y}_i$ are the gold and predicted classes, and $\mathbf{1}[\cdot]$ is the indicator function. For packet-level risk, y_i^{stat} and $\hat{y}_i^{\text{stat}}$ are the gold and predicted review statuses; $\mathcal{U}$ contains “insufficient visual evidence” and “not applicable without a visual fact”; and $\hat{\mathcal{L}}_i$ is the set of predicted clause links. The indicator u_i equals 1 exactly when $y_i^{\text{stat}} \in \mathcal{U}$ and either $\hat{y}_i^{\text{stat}}$ is “supported” or $|\hat{\mathcal{L}}_i| > 0$; otherwise, it equals 0. Here, $|\cdot|$ denotes set cardinality. The mean of u_i is reported as Unsupported-Link Rate (ULR) in the main comparison and as False-Supported Rate (FSR) in the status-stratified analysis.

Finally, r_i is the rank of the gold clause and k is the retrieval cutoff; hence, $\text{R@}k$ is recall at rank k , and MRR is mean reciprocal rank. The symbols $\leq$ and $\sum$ denote “less than or equal to” and summation, respectively.

The evidence layer reports $\bar{F}_1$ over localized evidence-label sets as Evidence Localization. The fact layer reports $\bar{F}_1$ over normalized fact tokens as Fact F1. The rule layer reports Acc for selected gold clauses as Rule Link Accuracy, together with Clause Relevance and ULR. The review-note layer includes JSON Validity, Note Factuality, and FSR. The oversight layer reports the presence rate of a non-empty human-verification field or an equivalent explicit textual flag; it is not a gold-label accuracy. The cross-page Consistency F1 is specifically $\bar{F}_1$ over normalized tokens in the generated and gold cross-page findings; cross-page status accuracy is reported separately. This distinction prevents Consistency F1 from being interpreted as a status-classification F1 score.

Overview score. Figure 1 uses one equal-weight summary only for a compact overview:

$$S_{\text{audit}}(m) = \frac{1}{5} (S_{\text{EL}}^{(m)} + S_{\text{FE}}^{(m)} + S_{\text{RL}}^{(m)} + S_{\text{PE}}^{(m)} + S_{\text{CP}}^{(m)}). \tag{3}$$

In Eq. (3), m identifies the evaluated model and $S_{\text{audit}}(m)$ is its unit-interval overview score. The five terms $S_{\text{EL}}^{(m)}$, $S_{\text{FE}}^{(m)}$, $S_{\text{RL}}^{(m)}$, $S_{\text{PE}}^{(m)}$, and $S_{\text{CP}}^{(m)}$ denote Evidence Localization label F1, Fact Extraction token F1, Rule Link Accuracy, review-packet Evidence-Label F1, and cross-page finding-token F1, respectively. The factor $1/5$ assigns equal weight to the five components. The PE term is the figure’s Review Traceability proxy; it measures localized-evidence

labels in the packet, not validity of the complete evidence–fact–clause chain. Multiplying $S_{\text{audit}}(m)$ by 100 gives the displayed percentage. For PlanTrace-VL, the unrounded value is 71.8435%, shown as 71.8%. This overview score does not replace the component-level comparisons below.

Main Results

Table 2 reports component-level performance on the 300-sample-per-task evaluation. PlanTrace-VL shows its clearest gains on structured intermediate outputs that are central to the paper’s claim. It reaches 95.67 Evidence Localization and 92.89 Fact F1, outperforming the prompting baselines on these two review-support functions. It also obtains the highest Consistency F1 in the table at 19.66, although the absolute value remains low. This result indicates that cross-page reasoning remains difficult in the proxy setting. Regulatory clause linking results are more mixed. PlanTrace-VL reaches 95.67 Rule Link Accuracy, but larger prompting baselines such as Qwen2.5-VL-72B-Instruct and Qwen2.5-VL-32B-Instruct remain competitive or slightly higher on isolated Rule Link Accuracy. Unsupported-Link Rate also requires cautious interpretation. PlanTrace-VL reduces this rate relative to several strong baselines with high unsupported-link rates, but it is not the lowest value in the table. These patterns support a bounded conclusion: the main benefit of PlanTrace-VL is improved traceable intermediate work, not universal dominance on every isolated metric.

End-to-end packet quality. The final column of Table 2 evaluates a central risk in end-to-end proxy review packets: unsupported links between a generated finding and its supposed evidence or clause support. This metric is more review-facing than JSON validity alone. A fluent review note can be well formatted but still unreliable for review if it incorrectly states support. PlanTrace-VL reduces this unsupported-link rate to 19.67, below several strong prompting baselines, although the result should be read diagnostically rather than as evidence of autonomous review readiness. Low values for models with poor packet parseability should not be interpreted as evidence of reliable review behavior.

Error and Calibration Analysis

Table 3 translates PlanTrace-VL’s remaining metric failures into review-facing risks. Unlocalized or mismatched evidence means that a note may lack visual support. Wrong or missing regulatory clause links mean that an applicable zoning provision may be misidentified. False-supported findings are especially important in planning review because they can make an uncertain case appear resolved. Invalid proxy JSON prevents downstream inspection of trace fields. Cross-page finding mismatch shows that distributed evidence remains difficult to reconcile. These errors should be read as audit signals, not only as aggregate performance losses. They identify where the system still requires human oversight and where future work should strengthen the pipeline.

Evidence-type difficulty. Table 4 stratifies the layout-patch grounding task by visual evidence type. Aggregate scores

can hide whether performance is stable across the categories that planners inspect in maps and drawings. In the current proxy evaluation, many structural drawing elements receive high localization scores, including Wall, Axis & Grid, Concrete Beam, Foundation, Glass, Steel Beam, Parking Space, Rebar, and Sink. Smaller or more visually ambiguous elements, such as Hole and Pile, show lower evidence scores. These evidence types should be interpreted as proxy visual categories for testing localization and traceability, not as a complete representation of urban planning judgment. This matters for planning review because strong performance on large structural elements does not guarantee reliable grounding of small symbols, parcel annotations, overlay boundaries, frontage labels, or setback-related markings.

Status calibration. Table 5 examines whether PlanTrace-VL preserves the human-verification pathway, a central safeguard in planning review. A cautious assistant that escalates uncertain cases is preferable to one that falsely states support without evidence. The model achieves high Status Accuracy for “Requires human verification”, reaching 98.90, but performs poorly on “Insufficient visual evidence”, where Status Accuracy is 0.00 and False-Supported Rate is 49.58. It therefore preserves escalation more reliably than it distinguishes insufficient evidence from other verification cases. This behavior reinforces the need for human-in-the-loop use and should not be interpreted as autonomous approval capability.

Retrieval Ablation and Limitations

Table 6 reports the rule-retrieval ablation used before response generation. Hierarchical retrieval improves the rank of the relevant planning provision relative to the flat Okapi Best Matching 25 (BM25) baseline: recall at rank 1 (Recall@1) increases from 94.70 to 97.93, and mean reciprocal rank (MRR) increases from 96.82 to 98.96. This improvement matters because cited clauses must be easy for planners to verify. In zoning and development review, a retrieved provision is a candidate clause rather than a final legal determination, and it should be inspected alongside the extracted fact and localized evidence.

Figure 6 visualizes capability profiles and retrieval-rank behavior without repeating the raw table entries. The figure supports the diagnostic interpretation of the experiments: the framework is evaluated by its ability to support planner verification through localized evidence, organized facts, surfaced clauses, and escalation of uncertain cases. The visualization complements the tables by showing capability balance and retrieval behavior rather than adding another model ranking.

Evidence status and limitations. The experiments are based on UrbanPlanTrace-Review and its proxy review tasks. They support the claim that PlanTrace-VL improves several auditable intermediate outputs, especially evidence localization, planning fact extraction, and traceable packet generation. They do not establish readiness for municipal deployment, full legal compliance checking, or autonomous planning approval. Because the visual splits include substantial technical drawing and CAD-style evidence, the

Table 2: Component-level performance on the reported 300-sample-per-task evaluation. Values are percentages. Higher is better except unsupported-link rate.

Model	Evidence Localization	Fact F1	Rule Link Accuracy	Consistency F1	Unsupported-Link Rate ↓
Mistral-Small-3.2-24B-Instruct-2506	73.33	62.58	93.33	2.14	24.00
MiniCPM-V-2.6-8B	64.67	63.27	19.00	0.42	39.67
Gemma-3-27B-IT	59.67	56.08	90.67	1.33	0.00
Gemma-3-12B-IT	43.00	55.05	91.33	1.21	0.00
Gemma-3-4B-IT	37.67	10.48	41.00	2.75	0.33
Qwen3-VL-32B-Instruct	42.67	37.42	0.00	1.40	0.00
Qwen3-VL-30B-A3B-Instruct	53.00	48.58	5.00	2.67	0.00
Qwen3-VL-8B-Instruct	62.00	50.95	0.00	1.06	0.00
Qwen3-VL-4B-Instruct	1.00	0.73	0.00	0.25	0.00
Qwen2.5-VL-32B-Instruct	58.67	73.33	96.33	1.28	39.00
Qwen2.5-VL-72B-Instruct	81.67	58.18	96.67	8.56	39.67
Qwen2.5-VL-7B-Instruct	93.00	61.98	90.67	11.89	39.67
Qwen2.5-VL-3B-Instruct	66.33	81.02	69.00	8.42	39.33
PlanTrace-VL	95.67	92.89	95.67	19.66	19.67

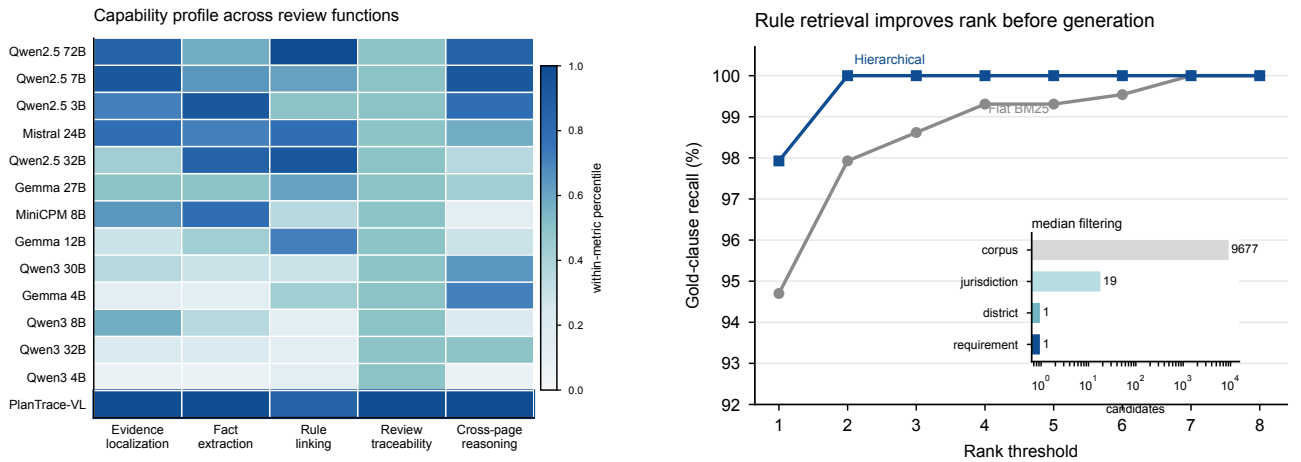


Figure 6: Experiment-level diagnostics for auditable planning document review. Left: capability profile across review functions, shown as within-metric percentiles. Right: gold-clause recall by retrieval-rank threshold, with an inset showing hierarchical filtering from the rule corpus to requirement-level candidates. B denotes billions of parameters, BM25 denotes the Okapi Best Matching 25 ranking function, and latency is measured in seconds (s).

Table 3: **PlanTrace-VL diagnostic errors.** Rates are percentages derived from the reported metrics; lower is better.

Failure mode and diagnostic	Rate (%) ↓	Review consequence
Evidence-localization error <i>100 – proxy Evidence F1</i>	44.67	Visual support cannot be traced
Clause-linking error <i>100 – proxy Clause Accuracy</i>	52.00	Applicable clause may be misidentified
False-supported finding <i>Proxy False-Supported Rate</i>	19.67	Unsupported output is labeled as supported
Missing verification flag <i>100 – Human Verification</i>	0.00	Verification flag is preserved in this run
Invalid trace JSON <i>100 – proxy JSON Validity</i>	1.00	Trace fields are not machine-parseable
Cross-page finding mismatch <i>100 – cross-page Finding F1</i>	80.34	Distributed evidence is not reconciled

results should be interpreted as diagnostic evidence about auditable grounding rather than as a complete representation of municipal review practice. Future work should validate the framework with real planning review packets,

Table 4: **Evidence-type breakdown.** Layout-patch grounding scores from the PlanTrace-VL prediction file; n is the sample count.

Evidence type	n	Score (%) ↑		
		Target Accuracy	Evidence F1	Fact F1
Concrete Column	50	92.00	92.00	90.68
Wall	49	100.00	100.00	98.75
Axis & Grid	36	100.00	100.00	88.00
Concrete Beam	36	100.00	100.00	93.65
Foundation	13	100.00	100.00	90.97
Hole	13	84.62	84.62	89.35
Pile	13	100.00	84.62	92.64
Glass	11	100.00	100.00	98.60
Steel Beam	10	100.00	100.00	95.14
Parking Space	9	100.00	100.00	100.00
Rebar	8	100.00	100.00	94.23
Sink	8	100.00	100.00	97.50

jurisdiction-specific rules, planner-in-the-loop evaluation, and more diverse document types, including zoning maps, parcel-level diagrams, site plans, permit drawings, legends, annotations, frontage conditions, and dimensional controls.

Table 5: **Review-status calibration.** Performance by gold review status; n is the sample count.

Gold status	n	Score (%)			
		Status Accuracy $\uparrow$	Evidence F1 $\uparrow$	Clause Accuracy $\uparrow$	False support $\downarrow$
Human verification required	181	98.90	58.01	46.41	0.00
Insufficient visual evidence	119	0.00	51.26	50.42	49.58
Overall	300	59.67	55.33	48.00	19.67

Table 6: **Rule-retrieval ablation.** Results on $n = 434$ hard clause-linking cases; gain is the absolute difference between hierarchical retrieval and the Okapi Best Matching 25 (BM25) baseline. R@1, R@3, and R@8 denote recall at ranks 1, 3, and 8; MRR denotes mean reciprocal rank.

Retriever	Retrieval metric (%) $\uparrow$			
	R@1	R@3	R@8	MRR
Flat BM25	94.70	98.62	100.00	96.82
Hierarchical retrieval	97.93	100.00	100.00	98.96
<i>Absolute gain</i>	+3.23	+1.38	+0.00	+2.14

Discussion

A main implication of PlanTrace-VL is that multimodal AI for planning should be evaluated around review workflows, not only answer accuracy. Planning review is an accountability practice in which findings must be explained, checked, and justified. A planner may need to identify why a frontage condition matters, where a setback measurement appears, which clause is relevant, and why a case should be escalated rather than resolved. Traceability is therefore a first-order property of planning AI, not a presentation feature added after generation.

This framing clarifies the paper’s position within planning support research. PlanTrace-VL does not replace planning judgment with an algorithmic decision rule. It tests whether a vision-language model can make review artifacts more inspectable for planner verification. This distinction matters because planning decisions are shaped by local ordinances, discretionary procedures, public values, and institutional accountability. These conditions are not fully captured by a proxy benchmark, so the results should be interpreted as evidence about auditable review support rather than municipal deployment readiness.

The results are most meaningful where planning review requires structured intermediate outputs. PlanTrace-VL improves evidence localization and planning fact extraction, and it produces review packets with explicit trace fields. These gains are practice-facing because they help organize the materials that planners already need to inspect: localized evidence, extracted facts, linked regulatory clauses, and auditable notes. The value of PlanTrace-VL therefore lies in making the review record more inspectable, while leaving interpretation, discretion, and legal responsibility with human planners.

The remaining errors define important research boundaries.

Unsupported findings, missing or wrong regulatory clause links, and cross-page mismatches are serious in planning review because they can make an uncertain record appear resolved. These errors are not merely technical failures; they are audit signals that show where professional verification remains necessary. For this reason, PlanTrace-VL should be understood as an evidence-grounding assistant that surfaces candidate evidence and candidate regulatory clauses, not as a substitute for professional code interpretation or public decision-making. Future validation should therefore prioritize real review packets, jurisdiction-specific rules, and planner-in-the-loop assessment, rather than treating larger models or larger benchmarks as sufficient on their own.

Conclusion

We present an auditable PlanTrace-VL workflow that links localized evidence and extracted facts to candidate clauses, structured notes, and human-verification status. UrbanPlanTrace-Review provides the proxy evaluation setting, and two-stage adaptation trains the model to produce this chain.

PlanTrace-VL achieves 95.67% Evidence Localization and 92.89% Fact F1, showing that domain adaptation can strengthen two intermediate outputs that planners need to inspect. The same evaluation also exposes material weaknesses: Consistency F1 remains 19.66%, the unsupported-link rate is 19.67%, and insufficient visual evidence is poorly calibrated. These results support a bounded conclusion: PlanTrace-VL should be interpreted as an evidence-grounding assistant, not an autonomous approval system, definitive legal compliance checker, or substitute for professional judgment.

The principal limitation is external validity: public proxy resources overrepresent technical drawings relative to municipal packets. Future work should evaluate jurisdiction-specific rules, diverse documents, cross-page aggregation, and planner-in-the-loop review on authentic case packets. Such evaluation should report retrieval success, evidence attribution, and downstream note factuality separately, because a correct candidate page does not guarantee a supported planning finding. This decomposition also prevents strong end-to-end scores from obscuring missed evidence, weak provenance, or unsupported intermediate links that would hinder professional review. Useful planning AI should make evidence easier to inspect and challenge, not conceal it behind fluent answers.

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